



TEAM : 8BIT

Sujit Wagh

Vaibhav Pokale

Yash Bhale

Raj Patil

Onkar Waghmode

PROBLEM STATEMENT

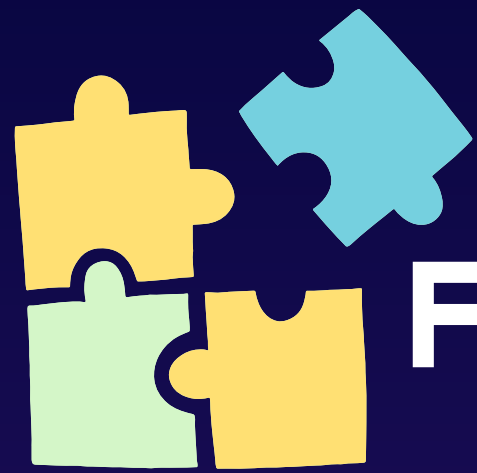
City-Wide Dark Store Network Projection

Design a projection model to determine the ideal number, size, and location of Blinkit dark stores within a city, factoring in customer demand patterns, delivery time constraints, and operational costs



SOLUTION OVERVIEW

- Customer demand analysis: Using clustering techniques to group high-demand locations.
- Capacity-aware store allocation: Predefining store sizes (small, medium, large) to distribute demand effectively.
- Traffic-aware placement: Factoring in road conditions to locate stores in easily accessible areas.
- Interactive visualization: Mapping clusters, demand heatmaps, and store locations for decision-making.



PROPOSED FEATURES

- Optimal dark store placement using clustering algorithm
- Geographical clustering of demand points
- Demand heatmap generation based on historical data
- Cost-based analysis (Rent, Petrol, Operational Costs)
- Expenditure report generation
- Store expansion to new cities

TECH STACK

Frontend: NextJS

Backend: Flask

Model: ML Algorithms

APIS: Leaflet, Google Traffic API

Dataset: Kaggle

Data Preprocessing: Python

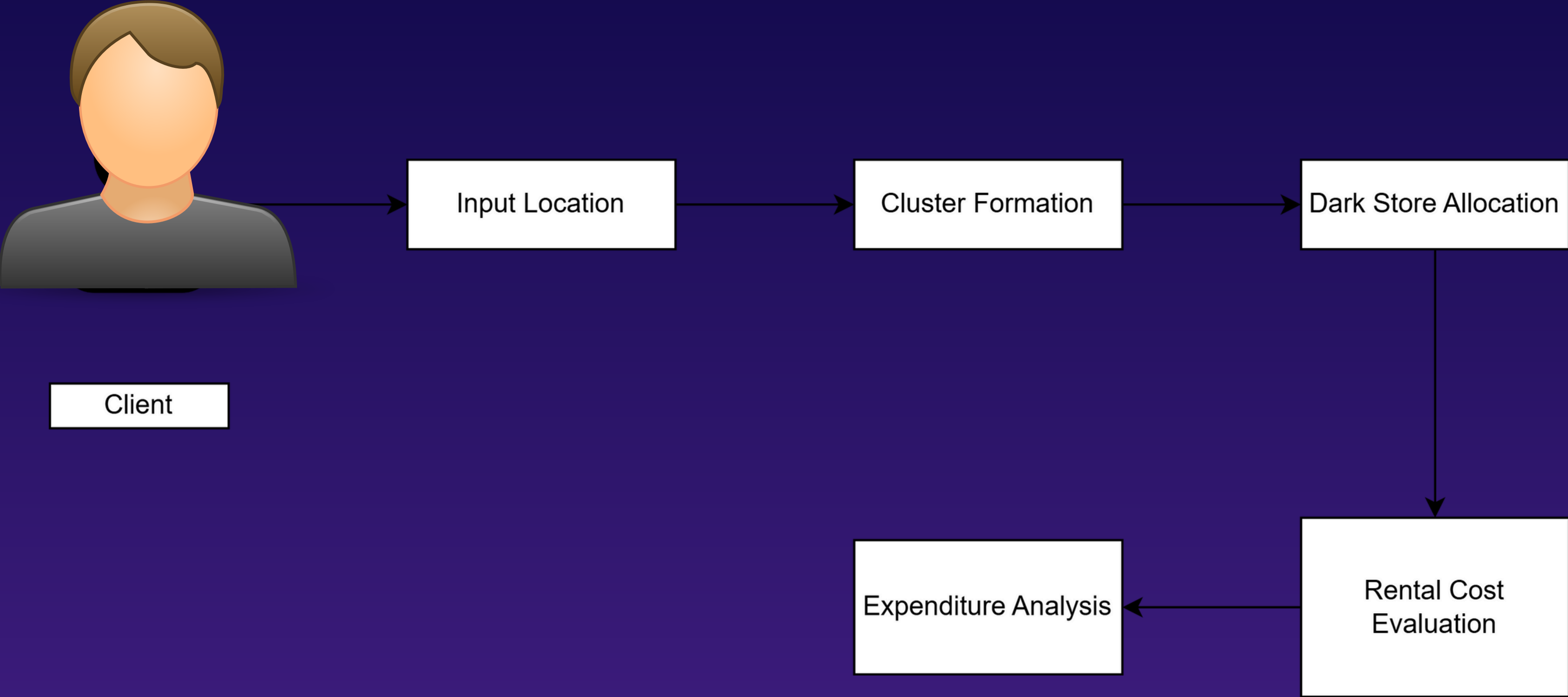


DATA RESOURCES REFERRED:

- Zomato Dataset for Traffic Demand
- Auto-Generated Dataset for Rental Analysis
- Mixture of Kaggle and Manual Dataset



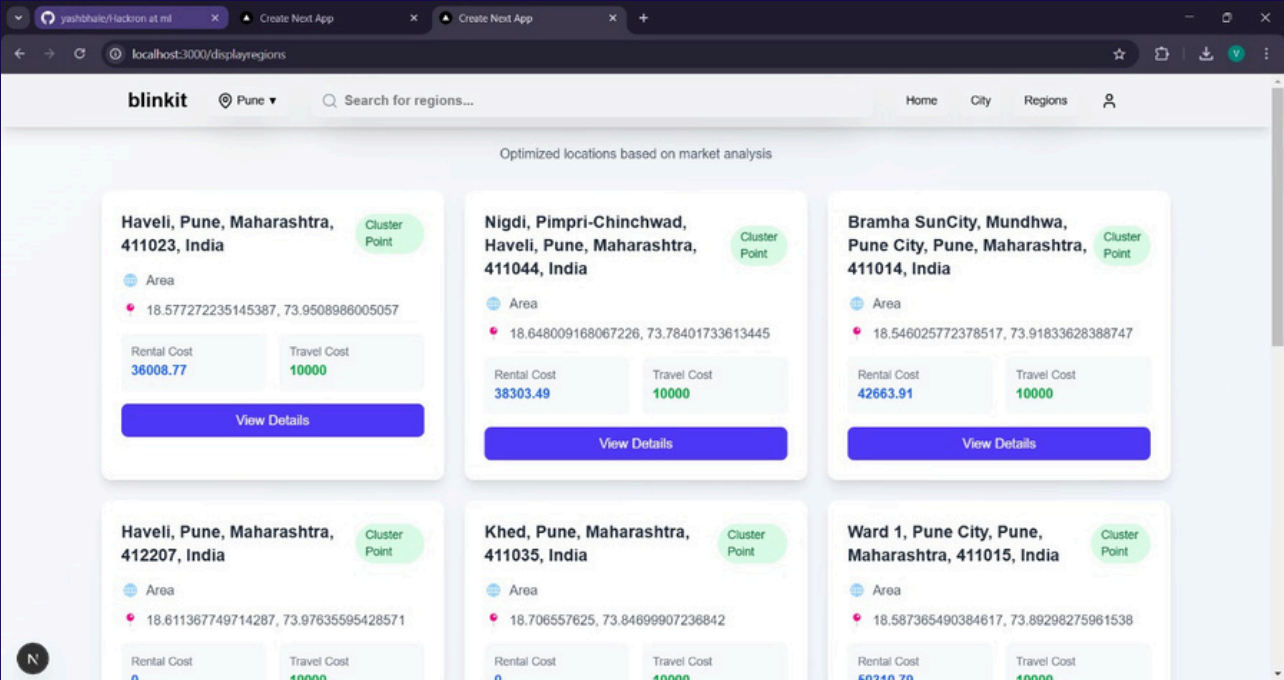
SOLUTION ARCHITECTURE OR USER DIAGRAMS



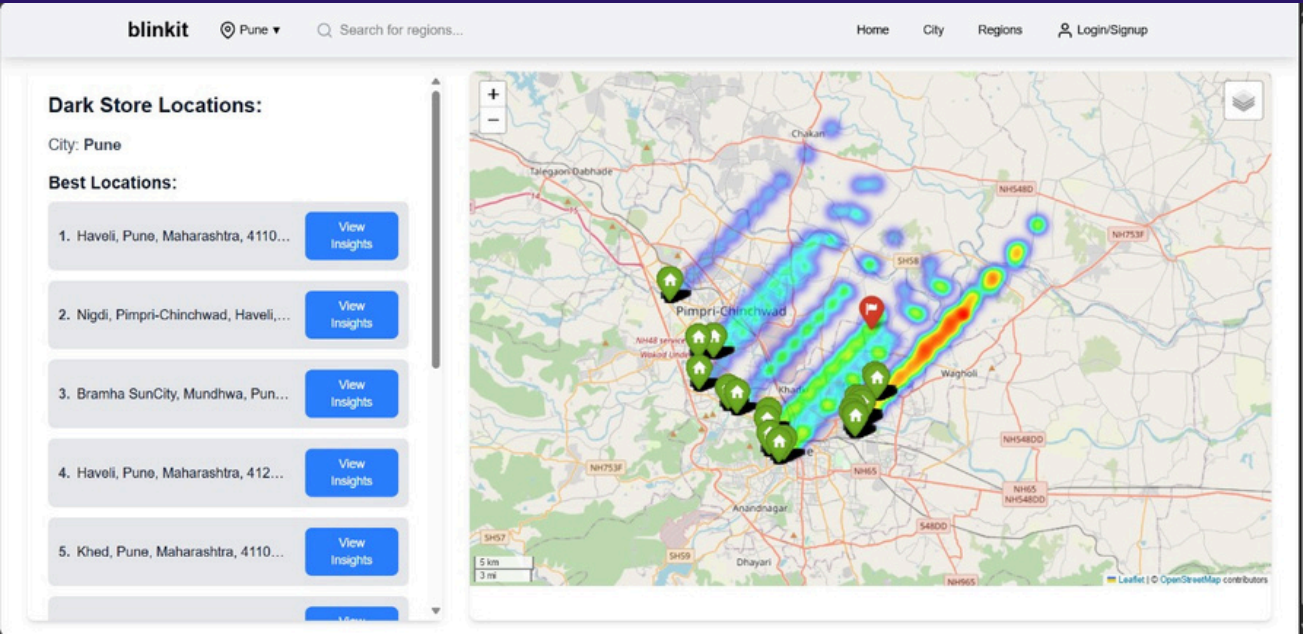
RESULTS:



Optimal dark stores locations



Location wise Expenditure cost



On demand Stores Expansion

FUTURE SCOPE & OUTCOMES EXPECTED

- 🚀 Real-Time Optimization – Use live traffic and demand fluctuations for dynamic store adjustments.
- 🚀 AI-Powered Demand Prediction – Implement machine learning to forecast future order patterns.
- 🚀 Delivery Route Optimization – Suggest best delivery routes considering congestion.
- 🚀 Automated Decision-Making – Develop an intelligent system to auto-recommend new store locations.



THANKS